

In the interaction shown in **Figure 54.1**, both organisms benefit: The cleaner wrasse gains access to a supply of food, and the moray eel is freed of parasites that might weaken it or spread disease. However, other interactions between members of different species can harm one of the participants, and still others can reduce the reproduction and survival of both participants. We'll begin this chapter by examining ecological interactions between the members of different species that live together in a community. We'll then turn to factors that affect **community structure**—the number of species found in a community, the particular species that are present, and the relative abundance of these species.

CONCEPT 54.1

Interactions between species can help, harm, or have no effect on the individuals involved

Some key relationships in the life of an organism are its interactions with individuals of other species in the community. These **interspecific interactions** include competition, predation, herbivory, parasitism, mutualism, and commensalism. In this section, we'll define and describe each of these interactions, grouping them according to whether they have positive (+) or negative (−) effects on the survival and reproduction of individuals engaged in the interaction.

For example, predation is a +/− interaction, with a positive effect on the survival and reproduction of members of the predator population and a negative effect on members of the prey population. Mutualism is a +/+ interaction in which the survival and reproduction of individuals of each species is increased in the presence of the other. A zero (0) indicates that the members of a species are not affected by the interaction.

We'll consider three broad categories of ecological interactions: competition (−/−), exploitation (+/−), and positive interactions (+/+ or +/0). As you study examples of these interactions, bear in mind that their effects can change over time. For example, an interaction that typically benefits individuals of both species may at times continue to benefit one of them, while harming or having no effect on the other.

Competition

Competition is a −/− interaction that occurs when individuals of different species each use a resource that limits the survival and reproduction of both individuals. (As described in Figure 53.18, competition for resources can also occur between members of the same species; this case is sometimes distinguished as *intraspecific competition*.) Weeds growing in a garden compete with garden plants for soil nutrients and water. Lynx and foxes in the northern forests of Alaska and Canada compete

for prey such as snowshoe hares. In contrast, some resources, such as oxygen, are rarely in short supply on land; most terrestrial species use this resource but do not usually compete for it.

Competitive Exclusion

What happens in a community when individuals of two species compete for limited resources? In 1934, Russian ecologist G. F. Gause studied this question using laboratory experiments with two closely related ciliate species, *Paramecium aurelia* and *Paramecium caudatum* (see Figure 28.19a). He cultured the species under stable conditions, adding a constant amount of food each day. When Gause grew the two species separately, each population increased rapidly in number and then leveled off at the apparent carrying capacity of the culture (see Figure 53.10a for an illustration of the logistic growth of a *Paramecium* population). But when Gause grew the two species together, *P. caudatum* became extinct in the culture. Gause inferred that *P. aurelia* had a competitive edge in obtaining food. More generally, his results led him to conclude that two species whose members compete for the same limiting resources cannot coexist permanently in the same place. In the absence of disturbance, one species will use the resources more efficiently and reproduce more rapidly than the other. Even a slight reproductive advantage will eventually lead to local elimination of the inferior competitor, an outcome called **competitive exclusion**.

Ecological Niches and Natural Selection

EVOLUTION Competition for limited resources can cause evolutionary change in populations. One way to examine how this occurs is to focus on an organism's **ecological niche**, the specific set of biotic and abiotic resources that an organism uses in its environment. The niche of a tropical tree lizard, for instance, includes the temperature range it tolerates, the size of branches on which it perches, the time of day when it is active, and the sizes and kinds of insects it eats. Such factors define the lizard's niche, or ecological role—how it fits into an ecosystem.

We can use the niche concept to restate the principle of competitive exclusion: Two species cannot coexist permanently in a community if their niches are identical. However, ecologically similar species *can* coexist in a community if one or more significant differences in their niches arise through time. Evolution by natural selection can result in one of the species using a different set of resources or similar resources at different times of the day or year. The differentiation of niches that enables similar species to coexist in a community is called **resource partitioning** (**Figure 54.2**).

As a result of competition, a species' *fundamental niche*, which is the niche potentially occupied by that species, is often different from its *realized niche*, the portion of its fundamental niche that it actually occupies. Ecologists can identify the fundamental niche of a species by testing the range of conditions in which it grows and reproduces in the absence of competitors. They can also test whether a potential